Interoperability Conformance Specification: hybridPrinterWithOverprintSimulation

Contents

Foreword

This document has been prepared following the ICC Intellectual Property Policy. This policy is based on the ITU-T/ITU-R/ISO/IEC Guidelines for Implementation of the Common Patent Policy (23 April 2012), with interpretations and clarifications to make it specific to ICC. A Patent Statement and Licensing Declaration form is available.

ICC Interoperability Conformance Specifications, of which this document is an example, may be submitted to the competent ISO Technical Committee for consideration and development as an ISO document. If so, this foreword is to be replaced by the appropriate wording supplied by ISO.

Introduction

ISO 20677-1 defines specifications that provide a platform for defining extended (iccMAX) colour management profiles and systems for various colour workflow domains. It provides a platform for which domain specific specifications can be defined that make use of iccMAX extensions to the existing cross-platform profile format of ISO 15076-1 (ICC) with greater flexibility for defining colour transforms and profile connection spaces to meet needs outside the scope of ICC profiles. However, not all systems provide support for iccMAX profiles thus limiting the ability to take advantage of extended capabilities at strategic points in the colour management ecosystem. Using a hybrid ICC profile that has an embedded iccMAX profile as a tag allows for compatibility with ICC systems that do not support iccMAX functionality while providing systems in the colour management ecosystem with iccMAX extensions to take advantage of extended colour management capabilities enabled by iccMAX profiles.

Requirements specifying restrictions to iccMAX that apply to a particular workflow are defined in workflow domain specifications known as Interoperability Conformance Specifications, of which this document is one example. Additionally, for some domain specific workflows it is envisioned that workflows will connect both to ICC profiles, iccMAX profiles, and hybrid profiles containing both ICC and iccMAX content.

An Interoperability Conformance Specification (ICS) is approved and registered by the International Color Consortium (ICC). It defines minimum structural and operational requirements for writing and reading iccMAX profiles in order to address a specific problem and/or functionality that cannot readily be handled using the ICC profile format defined by ISO 15076-1. An ICS document essentially defines restrictions to iccMAX content in profiles for specific use cases.

Interoperability Conformance Specification: overprintSimulation

1 Scope

This specification defines workflow requirements and restrictions for hybrid ICC profiles for printing that contain an iccMAX sub-profile as a tag for the purpose of providing color transforms with overprint simulation for additional colour channels not defined in the base profile. Overprint simulation profiles provide colorimetric transforms and optionally spectrally defined transforms.

The particular sub-set of the tags in the iccMAX sub-profle defined in ISO 20677-1 that are required to be present is defined, together with any optional tags that are permitted. The connections between profiles are described, and the processing elements that the CMM is required to support are identified.

2 Normative references

The following documents, in whole or in part, are normatively referenced in this document and are indispensable for its application. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO 15076-1:20##, *Image technology colour management — Architecture, profile format, and data structure*

NOTE: The most recent version of the ICC specification is available on the ICC web site [2].

ISO 20677-1:20##, *Image technology colour management — Extensions to architecture, profile format, and data structure: iccMAX*

NOTE: The most recent version of the iccMAX specification is available on the ICC web site [2].

3 Terms and definitions

All terms and definitions relevant to this document are provided in ISO 15076-1 and ISO 20677.

4 Use case

4.1 Domain

Profiles and workflows conforming to this ICS shall apply to the domain of colour management of printing processes including colour reproduction as well as previewing/proofing printed output.

4.2 Intended use

The intended use of this specification is to define workflows in which either colour data is converted to ink amounts for printing or ink amounts for printing are converted to another colour data representations for proofing, display, or other re-purposing aims. The ICC profile portion of a hybrid overprint printer profile (defined by ISO 15076-1) provides transforms between ink amounts and a colorimetric representation. The iccMAX portion of a hybrid overprint printer profile (defined by ISO 20677-1) provides transforms between ink amounts for additional channels as well as those provided by the ICC profile portion. Colorimetric transforms are required. Additional spectrally based and or

BRDF based transforms may also be provided. Because additional device ink channels are defined, additional metadata information about extra channels are also provided in the iccMAX portion. Spectral measurements of tints of the extra color channels can also optionally be provided with measurements over white and black backing.

4.3 Restrictions

ISO 15076-1 provides full details for an ICC printer profile. This document defines a set of restrictions which apply to an iccMAX sub-profile when it is embedded into an ICC printer profile, as a component of a hybrid profile created for the specific use cases described above. Such restrictions include the sub-set of tags from ISO 20677-1 which are permitted in such an iccMAX sub-profile conforming to this document.

5 Workflow

5.1 Hybrid sub-profile sub-classes

The profile class of a hybrid iccMAX sub-profile shall match the class of the containing ICC profile.

A supporting CMM shall support the profile classes and sub-profile sub-classes identified in Table 1 for conformance with this ICS. All profile classes are defined in ISO 15076-1 and ISO 20677-1.

Table 1. Profile classes and subclasses defined by this ICS

Profile	Profile class	Sub-profile sub-class identifier	Class signature	Sub-class signature
P	printer	overprintSimulation	'prn '	'osim'
M	multiplexIdentification	overprintChannelSelection	'mid '	'osel'

5.2 Connection Scenarios

5.2.1 General

The ICC profile portion of a hybrid overprint printer profile is intended to be used as either a source profile or a destination profile by any system that supports ICC printer class profiles that connection x device ink channels with a colorimetric PCS .

The 'osim' overprintSimulation sub-profile in a hybrid overprint printer profile connects *x+n* input channels (where *x* corresponds to the number of channels in the ICC portion and n corresponds to additional ink channels that are involved in overprint simulation) to a colorimetric Profile Connection Space which can be used as a source profile for overprint simulation. The colorspace of the ICC portion shall be a proper subset of the colorspace of the overprintSimulation subProfile.

Additionally, an overprintSimulation profile can optionally provide tags and header entries that enables MCS connection for flexible connection to additional channels

Tint measurement information over white and black can be provided for extra channels as additional metadata information about the extra color channels in an overprintSimulation sub-profile.

5.2.2 Profiles for ICS workflow scenarios

Table 2 provides overview information about additional profiles that are used to describe several profile connection scenarios conforming to this ICS.

Table 2. Additional profiles referenced in workflow scenarios defined by this ICS

Profile	Description
C	Profile conforming to ISO 20677-1 or ISO 15076-1 containing (a) colorimetric transform(s) with CMM configured to use a colorimetric transform type. If a separate ICS specification is associated with this profile it should be followed in addition to the connection requirements of this document.
1	Profile conforming to ISO 20677-1 containing Profile Connection Condition tags (spectralViewingConditionsTag, customToStandardPccTag, standardToCustomPccTag)
S	Profile conforming to ISO 20677-1 containing (a) spectral reflectance transform(s) with the CMM configured to use a spectral transform type. If a separate ICS specification is associated with this profile it should be followed in addition to the connection requirements of this document.

5.2.3 Workflow Scenarios

5.2.3.1 General

The following sections define scenarios that conform to this ICS.

Note: systems that only support ICC profiles will only work with scenarios 5.2.3.2 and 5.2.3.3.

5.2.3.2 Scenario 1: Connecting a source profile to a hybrid printer profile as destination with a colorimetric PCS

Profile Sequence	Profile Setup
 	C — Rendering Intent: Any Transform Type: Colorimetric PCC Override: None Use iccMAX Sub-profile: - P — Rendering Intent: Any Transform Type: Colorimetric PCC Override: None Use iccMAX Sub-profile: No

This workflow may be referred to as the “printing” workflow as it uses the base profile transforms in the hybrid overprint printer profile to generate data ready for printing using the base profile colorspace. The iccMAX sub-profile is not used for profile P, the base profile is used instead. This is the default behaviour of systems that do not support iccMAX profiles when this profile is used as a destination profile.

5.2.3.3 Scenario 2: Connecting a hybrid printer profile as source to a destination profile with a colorimetric PCS

Profile Sequence	Profile Setup
 	P — Rendering Intent: Any Transform Type: Colorimetric PCC Override: None Use iccMAX Sub-profile: No C — Rendering Intent: Any Transform Type: Colorimetric PCC Override: None Use iccMAX Sub-profile: -

This workflow may be referred to as a base “proofing” or “simulation” workflow as it uses the base profile transforms in the hybrid printer with reflectance profile to convert print ready data in the base profile color space to colorimetry for display or output. The iccMAX sub-profile is not used for profile P, the base profile is used instead. This is the default behaviour of systems that do not support iccMAX profiles when this profile is used as a source profile.

5.2.3.4 Scenario 3: Connecting an extended colour space with a hybrid overprint printer profile as source to a destination profile with a colorimetric PCS

Profile Sequence	Profile Setup
 	P — Rendering Intent: Any Transform Type: Colorimetric PCC Override: None Use iccMAX Sub-profile: Yes C — Rendering Intent: Any Transform Type: Colorimetric PCC Override: None Use iccMAX Sub-profile: -

This workflow may be referred to as an “overprint proofing” or “overprint simulation” workflow as it applies an extended ink set with the hybrid profile transforms in the hybrid printer profile to convert extended ink set print ready data to colorimetry for display or output. The iccMAX sub-profile is used for profile P, and an extended colour space is provided to profile P (which is a superset of the colour space of the base profile).

5.2.3.5 Scenario 4: Connecting an extended colour space with a hybrid overprint printer profile as source to a destination profile with a colorimetric PCS

Profile Sequence	Profile Setup		
	M	Rendering Intent:	Any
		Transform Type:	MCS
		PCC Override:	None
		Use iccMAX Sub-profile:	-
	P	Rendering Intent:	Any
		Transform Type:	MCS+ Colorimetric
		PCC Override:	None
		Use iccMAX Sub-profile:	YES
	C	Rendering Intent:	Any
		Transform Type:	Spectral
		PCC Override:	
		Use iccMAX Sub-profile:	Yes

For this optional scenario an MToBxTag and the multiplexTypeArrayTag, in the iccMAX sub-profile shall be present.

This workflow may be referred to as a selective “overprint proofing” or “overprint simulation” workflow as it first applies print ready data to a multiplex identification profile M to select channels from an extended ink set that are passed into an MCS hybrid profile transform in the hybrid printer profile P to convert values from an extended ink set to colorimetry for display or output. MSC connection with profile M allows for a selectable ordering of print data ready channels to be used with the overprint simulation. The iccMAX sub-profile is used for profile P with an MCS selected extended colour space provided to profile P (which includes colour channels from the superset of the colour space of the base profile).

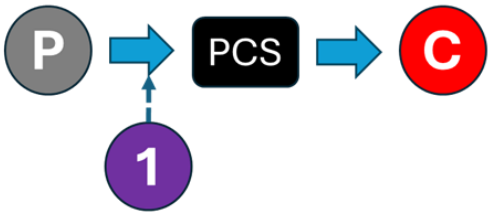
5.2.3.6 Scenario 5: Getting overprint tint colour information using a hybrid overprint printer profile as source with a PCC override to a destination profile with a colorimetric PCS

Profile Sequence	Profile Setup		
	P	Rendering Intent:	Any
		Transform Type:	Named Color to Spectral
		PCC Override:	None
		Use iccMAX Sub-profile:	Yes
	C	Rendering Intent:	Any
		Transform Type:	Colorimetric
		PCC Override:	None
		Use iccMAX Sub-profile:	-

For this optional scenario the NamedColor tag in the iccMAX sub-profile shall be present.

This workflow may be referred an overprint colour tint query as it uses the NamedColor tag in the iccMAX sub-profile transforms look up colorimetric PCS values for tints of named color channels in the extended color space of profile P. This PCS data is converted using the PCC override in profile 1 to get colorimetry passed to profile C for display or output. Both the name of the channel and tint value are provided to an iccMAX sub-profile named colour transform to get colorimetric or spectral reflectance PCS data.

5.2.3.7 Scenario 6: Getting overprint tint colour information using a hybrid overprint printer profile as source with a PCC override to a destination profile with a colorimetric PCS

Profile Sequence	Profile Setup
	P Rendering Intent: Any Transform Type: Named Color to Spectral PCC Override: 1 Use iccMAX Sub-profile: Yes C Rendering Intent: Any Transform Type: Colorimetric PCC Override: None Use iccMAX Sub-profile: -

For this optional scenario the NamedColor tag in the iccMAX sub-profile shall be present.

This workflow may be referred an overprint colour tint query as it uses the NamedColor tag in the iccMAX sub-profile transforms look up spectral PCS values for tints of named color channels in the extended color space of profile P. This PCS data is converted using the PCC override in profile 1 to get colorimetry passed to profile C for display or output. Both the name of the channel and tint value are provided to an iccMAX sub-profile named colour transform to get colorimetric or spectral reflectance PCS data.

5.2.3.8 Scenario 7: Getting overprint tint colour information using a hybrid overprint printer profile as source with a PCC override to a destination profile with a colorimetric PCS (used for spot ink formulation)

Profile Sequence	Profile Setup
	P Rendering Intent: Any Transform Type: Named Color to Spectral PCC Override: None Use iccMAX Sub-profile: Yes S Rendering Intent: Any Transform Type: Colorimetric PCC Override: None Use iccMAX Sub-profile: -

For this optional scenario the NamedColor tag in the iccMAX sub-profile shall be present.

This workflow may be referred an overprint colour tint query. The main purpose for this scenario is to allow the spectral measurements of the spots to be determined for spot ink formulation allowing for spectrally correct printing using the formulateds spot colors in a press. This scenario uses the

NamedColor tag in the iccMAX sub-profile transforms look up of spectral PCS values for tints of named color channels in the extended color space of profile P. This PCS data is to profile S to output as a spectral encoding. Both the name of the channel and tint value are provided to an iccMAX sub-profile named colour transform to get spectral reflectance PCS data.

6 Sub-class Profile Requirements

6.1 General

Requirements for hybrid iccMAX profiles conforming to this ICS document are listed here. ICC v4 profiles shown in Section 5 above shall conform to ISO 15076-1.

6.2 Requirements

The requirements for a hybrid overprint printer profile with reflectance are in two parts.

The encoding of the base profile shall be as defined in ISO 15076-1 as a printer class profile.

An extension iccMAX sub-profile shall also be encoded in this base (encapsulating) profile as defined in the embedding annex in ISO 20677-1. This involves adding an embeddedV5ProfileTag (signature ‘ICC5’) to the base profile, with the tag encoded as an embeddedProfileType (signature ‘ICCP’) with the sub-profile encoded as defined in ISO 20677-1.

The encoding of the profile header of the encapsulated sub-profile in the embeddedV5ProfileTag shall be as defined in ISO 20677-1, with the specific requirements shown in Table 3.

Table 3. Header requirements

Header field	Required content
Profile version	‘5.0’ or later. Where the spectral PCS field is non-zero and the observer CMF and illuminant SPD are omitted from the spectralViewingConditionsTag the profile version shall be ‘5.1’ or later.
Profile class	‘prn ’
Profile subclass	‘osim’
Profile subclass major version	1
Profile subclass minor version	0
Profile flags	Embedded
Device attributes	
Data colour space	‘ncxxxx’ where xxxx refers to number of Data colour channels. The first channels of the color space shall be the same as in the encapsulating ICC profile
MCS	0 (when no MToB0 tag exists) or “mcxxxx’ where xxxx refers to number of MCS channels (when MToB0 tags exist). The number of MSC channels shall be the same as the number of channels in the Data color space.
Colorimetric PCS	‘Lab ‘ or ‘XYZ ‘
Spectral PCS	0 or ‘rsxxxx’ where xxxx is a hex value corresponding to the number of spectral channels in the PCS (associated with spectral measurements in a namedColorTag if it exists and has spectral data)

Spectral PCS range	Start and end wavelengths of the spectral PCS encoded as float16 values
Bispectral PCS	0

Full details of the encoding of the header fields in 3 are given in ISO 20677-1.

6.2.1 Required tags

Profiles shall contain the tags listed in Table 4.

Table 4. Required tags

Tag name	Signature	Required content
colorantInfoTag	'clin'	Information about each of the channels in the extended ColorSpace
AToB0, AToB1, AToB2, or AToB3	'A2B0', 'A2B1', 'A2B2', or 'A2B3'	multiProcessElementType element containing zero or more curveSetElement, matrixElement, CLUTElement, extendedCLutElment, calulatorElement, or tintElement processing elements

The encoding of the tags listed in Table 4 shall be as defined in ISO 20677-1.

6.2.2 Optional tags

Profiles may contain the tags listed in Table 5.

Table 5. Optional tags

Tag name	Signature	Required content
Other AToB0, AToB1, AToB2, or AToB3 tags	'A2B0', 'A2B1', 'A2B2', or 'A2B3'	multiProcessElementType element containing zero or more curveSetElement, matrixElement, CLUTElement, extendedCLutElment calulatorElement, or tintElement processing elements
Any or none of MToB0, MToB1, MToB2, or MToB3 tags	'M2B0', 'M2B1', 'M2B2', or 'M2B3'	multiProcessElementType element containing zero or more curveSetElement, matrixElement, CLUTElement, extendedCLutElment calulatorElement, or tintElement processing elements. The number of channels shall match the MCS in header. (Note: the tag data for MToBx and AToBx tags may be shared)
multiplexTypeArrayTag	'mcta'	tagArray of utf8Type with tagArray type signature of 'mcta' that defines multiplex channel type names for each of the channels in the MCS. This tag shall

		be present if one or more MToBx tags are present.
namedColorTag	'nclr'	A tagArrayType as a namedColorArray containing zero tint and named colors that should match colors found in the colorantInfoTag. If spectral information is in the namedColorTag then the spectral PCS information in the header shall be present and a spectralViewingConditionsTag shall also be present. Colorimetric information (if present) corresponds to the colorimetric PCS field in the header
multiplexDefaultValuesTag	'mdv '	uInt8ArrayType, uInt16ArrayType, float16ArrayType, or float32ArrayType. This tag may be present if a multiplexTypeArrayTag is present. If not present default values for missing channels shall be assumed to be 0.
spectralViewingConditionsTag	'svcn'	Structure defining the observer and illuminant associated with the spectral PCS. This tag shall be present whenever the spectral PCS field in the header is non-zero. For a profile version of '5.1' or later the observer CMF and illuminant SPD may be omitted when standard observer and standard illuminant identifiers are provided.
customToStandardPccTag	'c2sp'	multiProcessElementType element containing zero or more curveSetElement, matrixElement, CLUTElement, or extendedCLutElment processing elements that convert colorimetry based on the observer and illuminant defined by the spectralViewingConditionsTag to standard colorimetry. This tag shall be present when the spectral PCS field in the header is non-zero and the observer and illuminant defined by the spectralViewingConditionsTag are not the CIE 1931 standard 2 degree observer and the D50 illuminant.
standardToCustomPccTag	's2cp'	multiProcessElementType element containing zero or more curveSetElement, matrixElement, CLUTElement, or extendedCLutElment processing elements that perform the inverse of the transform in the customToStandardPccTag. This tag shall be present under the same conditions as the customToStandardPccTag.

The encoding of the tags listed in Table 5 shall be as defined in ISO 20677-1.

6.2.3 Channel naming guidance

Overprint simulation profiles extend the colorspace of the base profile. Names for the first channels in the sub-profile shall agree with channels in the base profile. Thus if the base profile uses a CMYK color space the first 4 channel names of the sub-profile shall be "Cyan", "Magenta", "Yellow", and "Black". If the base profile uses an N-color colour space then the names first colour channels in the sub-profile shall be the same as the names of the color channels in the base profile.

Additional channels correspond to spot colors. These names should match those of the images \ documents that the profile is used with.

6.2.4 Associated CMM environment variables

Overprint simulation profiles are likely implemented using a calculatorElement that allows for CMM environment variables to be used to enable runtime customization of the transform. Each CMM environment variable is associated with a 32-bit signature name and a floating point value. The CMM environment variable names and values in Table 6 are recommended for use in the implementation of overprint modelling.

Table 6. CMM Environment Variables

Signature	Value Encoding
'0ni?'	Flag to indicate if spot channels should not be inverted (photoshop inverts spot channels in CMYK images). (0 indicates invert spots, >0.5 indicates spot should not be inverted). If this ENV is not provided then spots are inverted.
'bkgX'	CIE X value of background colour when all device channels are 0. A default value will be used if this is not provided to the CMM.
'bkgY'	CIE Y value of background colour when all device channels are 0. A default value will be used if this is not provided to the CMM.
'bkgZ'	CIE Z value of background colour when all device channels are 0. A default value will be used if this is not provided to the CMM.

6.2.5 Example

The profile CMYK_overprintSim_printer.icc is encoded according to the requirements of this ICS document and is available in the iccDEV-Testing suite [3]. An XML representation is also provided.

7 Additional Profile Requirements

7.1 General

Requirements for additional iccMAX profiles conforming to this ICS document are listed here.

7.2 Requirements

An iccMAX overprint channels selection profile shall also be encoded as a multiplexIdentification class profile as defined in ISO 20677-1. The encoding of the profile header of the overprint channel selection profile shall be as defined in ISO 20677-1, with the specific requirements shown in Table 7.

Table 7. Header requirements

Header field	Required content
Profile class	'mid '
Profile subclass	'osel'
Profile subclass major version	1
Profile subclass minor version	0
Profile flags	Embedded
Device attributes	
Data colour space	0
MCS	"mcxxxx' where xxxx refers to number of MCS channels.
Colorimetric PCS	0
Spectral PCS	0
Spectral PCS range	0
Bispectral PCS	0

Full details of the encoding of the header fields in Table 7 are given in ISO 20677-1.

7.2.1 Required tags

Profiles shall contain the tags listed in Table 8.

Table 8. Required tags

Tag name	Signature	Required content
AToM0	'A2M0'	multiProcessElementType element containing zero or more curveSetElement, matrixElement, CLUTElement, extendedCLutElment, calculatorElement, or tintElement processing elements. Normally there are zero element for AToM0 tags in the transform unless processing is needed to convert the encoding .

The encoding of the tags listed in Table 8 shall be as defined in ISO 20677-1.

7.2.2 Example

The profile CMYK_overprintMid.icc is encoded according to the requirements of this ICS document and is available in the iccDEV-Testing suite [3]. An XML representation is also provided.

8 Conformance

A hybrid printer profile with output simulation shall be considered to be in conformance with this ICS document if it meets the following conditions:

- The profile connects to the channels specified in section 5.
- The profile header includes the required content from Table 3.
- All required tags listed in Table 4 are present in the profile.
- Tags listed in Table 5 whose required content states a condition are present in the profile whenever that condition applies.
- The profile structure and all tags conform to ISO 20677-1.

An iccMAX overprint channels selection profile shall be considered to be in conformance with this ICS document if it meets the following conditions:

- The profile connects to the channels specified in section 5.
- The profile header includes the required content from Table 7.
- All required tags listed in Table 8 are present in the profile.

- The profile structure and all tags conform to ISO 20677-1.

A CMM shall be considered to be in conformance with this ICS if it meets the following conditions:

- The CMM is able to parse profiles that conform to this ICS
- The CMM supports and is capable of processing the channels specified in section 5 for supported scenarios and any other profiles listed in Table 1 and Table 2.
- The CMM is able to process the tags listed in Table 4, Table 5, and Table 8 for scenarios that it supports. (Note: optional scenarios may not be supported by an implementation)
- When processing a profile conforming to this ICS, the CMM produces results that are a close approximation to those produced by the iccMAX Reference Implementation [3]

Bibliography

[2] iccMAX http://www.color.org/iccmax/

[3] iccMAX Reference Implementation http://www.color.org/iccmax/index.xalter#reficcmax

9 Annex A (informative) - Worked example scenarios

This annex describes the worked example scenarios provided in the iccDEV-Testing ICS package named HybridPrinterWithOverprintSimulation. Each entry is driven by a JSON configuration file in the package config directory and demonstrates one of the workflow scenarios defined in clause 5.2.3 applied to the example hybrid overprint printer profiles.

Table A.1 - Worked example scenarios in the iccDEV-Testing ICS package

ID	Clause 5.2.3 scenario	iccDEV tool / driver file(s)	What is demonstrated
S1	Scenario 1 (5.2.3.2)	iccApplyProfiles / hpwos-S1-PrintOutput.json	Rendering an embedded-sRGB image into CMYK through the base part of the hybrid printer profile (perceptual).
S2	Scenario 2 (5.2.3.3)	iccApplyProfiles / hpwos-S2-PrintProof.json	Base colorimetric (D50, absolute) proof of the CMYK output back to sRGB, not using the osim sub-profile.
S3	Scenario 3 (5.2.3.4)	iccApplyProfiles / hpwos-S3-TShirtDesignPrev{UW,US,OS}-{W,R,G,B,K}.json (15 files)	Overprint simulation of an extended CMYK+spot design via the osim sub-profile, over three spot configurations (under-white, under-silver, silver-on-top) and five substrate colours set through bkgX/Y/Z.
S4	Scenario 4 (5.2.3.5)	iccApplyProfiles / hpwos-S4-TShirtDesignPrev{UW-G-M,US-G-M,US-W-CS}.json (3 files)	Selective overprint simulation: an osel multiplexIdentification profile selects and orders extended ink channels over an MCS connection into the hybrid profile.
S5	Scenario 5 (5.2.3.6)	iccApplyNamedCmm / hpwos-S5-SpotTintSampleLab.json	Named-colour (spot) tint query returning colorimetric Lab values (D50).
S6	Scenario 6 (5.2.3.7)	iccApplyNamedCmm / hpwos-S6-SpotTintSamplePccLab.json	Named-colour (spot) tint query with a PCC override (D93).
S7	Scenario 7 (5.2.3.8)	iccApplyNamedCmm / hpwos-S7-SpotTintSampleRefOver{White,Black}.json (2 files)	Named-colour (spot) tint query returning spectral reflectance over white backing and over black backing. Used for getting spectral

			measurements for ink formulation.

The S1-S4 image scenarios write TIFF outputs to the Results directory; the S5-S7 named-colour queries write their results to Results/SpotTintSample*.txt. All driver files use POSIX-style forward-slash paths so the same configurations run under both BuildAndTest.bat and BuildAndTest.sh.